6G / NEXT-GENERATION WIRELESS

6G AND BEYOND :

The Prospects for Intelligent Wireless Connectivity

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Our route through 6G

Six parts: from the 6G vision to applications, challenges and conclusions.

01 SLIDE 03

Introduction

Vision and presentation scope

02 SLIDES 04–05

Background & technology evolution

From voice calls to IMT-2030

03 SLIDES 06–08

Core technologies of 6G

AI, radio, sensing and integrated coverage

04 SLIDE 09

Applications & case studies

Education, transport and renewable energy

05 SLIDES 10–11

Challenges & future research

Reliability, security and sustainability

06 SLIDE 12

Conclusion

Key messages and evidence-led outlook

References follow on slides 13–14.

What if the network could do more?

6G is not only about faster downloads — it is about systems that communicate, sense and adapt.

Imagine a virtual laboratory at home — immersive, interactive, and responsive.

Now imagine the wireless network also senses movement and adapts resources in real time.

01

CONNECT

Move information reliably between people, devices and services.

02

SENSE

Use radio signals to estimate location, movement and context.

03

ADAPT

Use intelligence to respond to changing conditions.

evolution → technologies → applications → challenges

[1], [2]

From voice calls to intelligent connectivity

Each generation expanded what mobile networks could support.



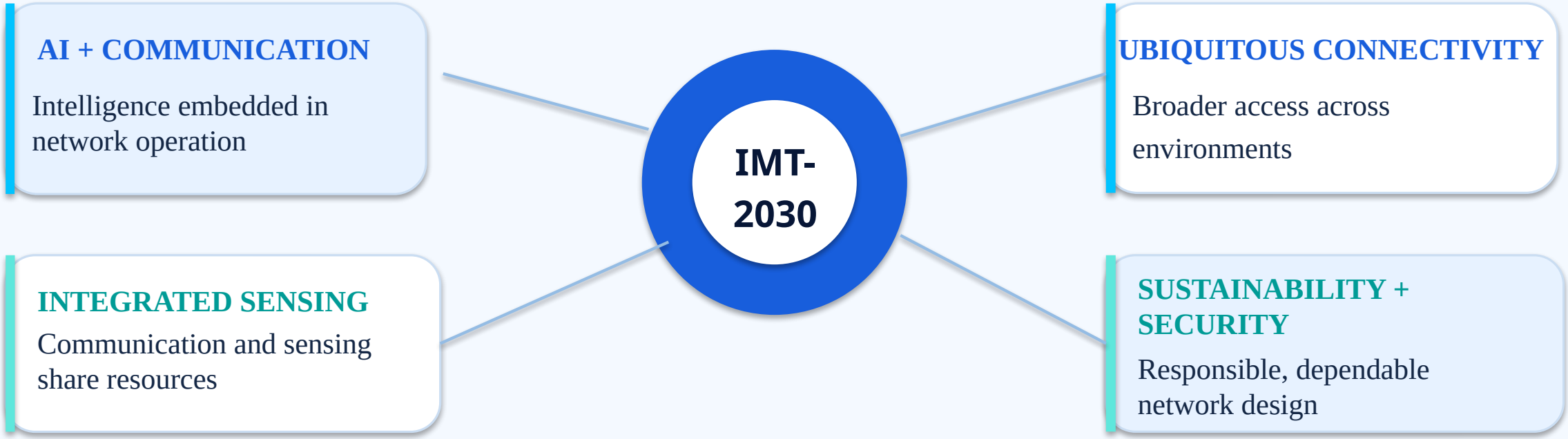
1G	2G	3G	4G	5G	6G
Analog voice	**Digital voice**	**Mobile internet**	**Mobile broadband**	**Service diversity**	**Converged systems**
Voice calls	Messaging	Web access	Everyday apps	High capacity • low delay • massive IoT	Communication + sensing + AI

The 6G question: how can multiple capabilities work together — reliably, securely and sustainably?

[1], [2]

IMT-2030: a broader ambition

The ITU framework looks beyond incremental speed improvements.



A framework defines goals — not a guarantee that every target will be achieved simultaneously.

[1]

Four technology building blocks

The 6G vision depends on complementary advances — each with its own constraints.

AI

01 AI-native networking

Predict traffic, allocate resources and maintain connections; edge computing can process data closer to users.

RF

02 New radio approaches

Sub-THz / THz bandwidth, large antenna arrays and reconfigurable intelligent surfaces.

ISAC

03 Integrated sensing & communication

Reuse radio signals and infrastructure to communicate while estimating location or movement.

NTN

04 Integrated coverage

Coordinate terrestrial networks with satellites and aerial platforms to extend connectivity.

[2]–[4]

Intelligence meets the radio environment

Two complementary levers: smarter network decisions and more capable radio links.

AI-NATIVE NETWORKING



Potential: traffic prediction • resource allocation • connection maintenance

Watch-out: training data, compute demand and decision reliability.

RADIO CAPACITY & PROPAGATION



Sub-THz / THz
- **large antenna arrays**
- **intelligent surfaces**

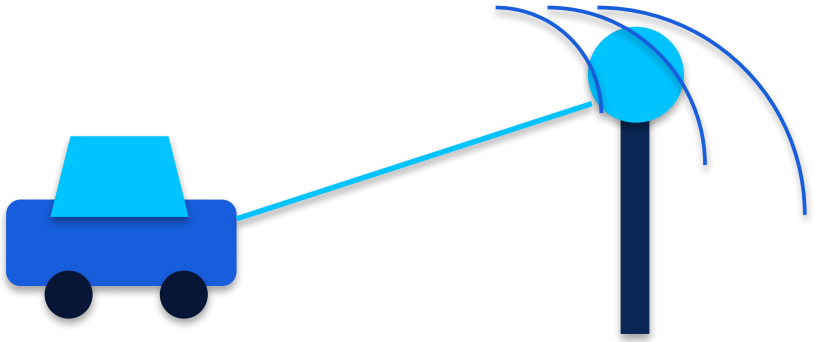
Watch-out: blockage, propagation loss and hardware limits. Higher bands complement lower-frequency coverage.

[2], [3]

One network, more awareness

ISAC and integrated coverage expand what a wireless system can observe and reach.

ISAC / SENSING + COMMUNICATION



Sense position / motion while supporting data exchange

Trade-off: sensing accuracy and communication performance may compete for resources.

INTEGRATED COVERAGE



Terrestrial networks + satellites + aerial platforms

Challenge: coordinate very different links while platforms and users move.

[2], [4]

Three ways 6G could matter

These are literature-based use cases and proposed designs — not completed commercial 6G deployments.

CASE 01

Immersive education

Virtual laboratories need high-quality visuals and consistent interaction. Edge processing and intelligent allocation could help.

Evaluate: interaction delay, interruptions, user comfort.

CASE 02

Connected transport

Roadside sensing could track vehicle motion, support narrow radio beams and prepare resources before handover.

Evaluate: sensing accuracy, connection reliability.

CASE 03

Renewable-energy coordination

Timely information from solar installations, batteries and consumers could help balance local supply and demand.

Evaluate: cost, power use, compatibility.

The test: what measurable benefit does the proposed design add over today's alternatives?

[2], [4]–[6]

Promising components must work in the real world

System-level evidence matters more than isolated component performance.

SYSTEM

01 Real-world reliability

Mobility, congestion, blockage and equipment constraints can change results.

Need: reproducible testbeds and fair baseline comparisons.

TRUST

02 Security & privacy

Manipulated data can mislead learning systems; sensing can expose sensitive movement information.

Need: built-in protections and adversarial testing.

IMPACT

03 Sustainability

Infrastructure, computing and hardware consume resources beyond the energy used to transmit bits.

Need: report energy, deployment and maintenance costs.

[2], [3], [7], [8]

Three questions to keep 6G grounded

A practical research agenda connects performance claims to real operating conditions.

01

RELIABILITY

Does the system remain reliable outside ideal conditions?

Test mobility, congestion, blockage and equipment limits.

02

RESILIENCE

Can it protect users and recover from attacks?

Test data / model attacks, privacy risks and recovery.

03

SUSTAINABILITY

Do the benefits justify energy and deployment costs?

Measure system-level energy use and lifecycle costs.

Progress requires collaboration across communications, computing, security and application design.

[2], [3], [7], [8]

The future of 6G must be demonstrated

A compelling vision is one supported by useful applications and realistic evidence.

Faster is not enough.

More intelligent is not automatically secure or sustainable.

BETTER EXPERIENCE

Useful applications and human-centered metrics

RELIABLE OPERATION

Evidence under realistic conditions

RESPONSIBLE RESOURCES

Security, energy and deployment trade-offs

Thank you for listening.

Selected sources

Sources cited in the presentation — original titles retained for traceability.

[1] International Telecommunication Union. “Framework and overall objectives of the future development of IMT for 2030 and beyond.” Recommendation ITU-R M.2160-0, Nov. 2023. itu.int/rec/R-REC-M.2160-0-202311-I/en

[2] S. Latreche and H. Bellahsene. “A comprehensive survey on 6G: Enabling technologies, key applications, and future challenges.” Franklin Open, vol. 15, Art. 100559, 2026. doi:10.1016/j.fraope.2026.100559

[3] K. B. Letaief et al. “The roadmap to 6G – AI empowered wireless networks.” arXiv:1904.11686v2, 2019. arxiv.org/abs/1904.11686v2

[4] F. Liu et al. “Integrated sensing and communications: Towards dual-functional wireless networks for 6G and beyond.” arXiv:2108.07165v1, 2021. arxiv.org/abs/2108.07165v1

Further reading

Full source details are provided in the original English script.

[5] W. Saad, M. Bennis and M. Chen. “A vision of 6G wireless systems: Applications, trends, technologies, and open research problems.” arXiv:1902.10265v1, 2019. arxiv.org/abs/1902.10265v1

[6] A. Rashid et al. “Review on 6G communication network technology for sustainable renewable energy system: Implementations, constraints, and recommendations.” Results in Engineering, vol. 28, Art. 107933, 2025. doi:10.1016/j.rineng.2025.107933

[7] V.-L. Nguyen et al. “Security and privacy for 6G: A survey on prospective technologies and challenges.” arXiv:2108.11861v2, 2021. arxiv.org/abs/2108.11861v2

[8] I. Ahmad et al. “Sustainability in 6G networks: Vision and directions.” Proc. 2023 IEEE Conference on Standards for Communications and Networking, pp. 202–208. doi:10.1109/CSCN60443.2023.10453200

Evidence-led overview • Proposed capabilities are development goals, not guarantees of deployment.